KOGANTI SASANK

Data Science Technical Lead and AI Solution Architect

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PROFESSIONAL SUMMARY

Senior Data Scientist and technical lead with 4+ years of experience architecting and delivering production AI, Generative AI, machine-learning and optimization systems. Leads solution design, project planning, engineering standards, mentoring and stakeholder delivery while remaining hands-on with Python, SQL, LangGraph, FastAPI, MLOps and cloud-native deployment. Delivered measurable business outcomes in industrial optimization and enterprise AI.

CORE SKILLS

Technical leadership: Solution architecture, project planning, requirements translation, design reviews, mentoring, hiring, stakeholder communication, guild leadership
AI and GenAI: LangGraph, multi-agent systems, agentic RAG, LLM evaluation, prompt engineering, Gemma, Llama 3.1, vLLM, Ollama, NeMo Guardrails
ML and optimization: XGBoost, scikit-learn, model validation, MILP, differential evolution, Gurobi, CPLEX, PuLP
Platform and MLOps: Python, SQL, FastAPI, REST APIs, Docker, Kubernetes, KEDA, RabbitMQ, Celery, Redis, Jenkins CI/CD, MLflow, Kubeflow, Langfuse, GCP

PROFESSIONAL EXPERIENCE

Senior Data Scientist | TCG Digital, Remote, India | Jul 2025 - Present

- Lead architecture and delivery for an enterprise procurement platform with a supervisor and five specialized agents, integrating SAP Ariba, email, databases, deterministic SQL routing and document-grounded RAG workflows.
- Designed production controls for structured agent inputs and outputs, restricted tool access, response evaluation, confidence scoring, guardrails, role-aware retrieval, observability and human review.
- Scaled asynchronous processing for up to 2,000 concurrent users using RabbitMQ, Celery, Redis, Kubernetes and KEDA; established Langfuse tracing and automated quality checks.
- Led a 50,000+ patent research engine using BM25, LLM scoring and QLoRA fine-tuning; achieved 65% relevance and deployed the workflow on GCP.
- Lead productization of a real-time refinery optimizer that delivered a client-reported 1.5%-1.8% yield uplift and approximately INR 3,000-4,000 per ton-hour margin improvement.
- Architected grade-market and production-sequencing optimization that generated approximately INR 2 crore in estimated business benefit.
- Manage two team members, mentor 5-6 colleagues, lead the Agentic Innovation Guild, co-lead the DevOps and Optimization guilds and have interviewed 10-12 GenAI candidates.

Data Scientist Consultant | TCG Digital, Remote, India | Jul 2022 - Jun 2025

- Built reusable CI/CD standards for data-science APIs using linting, Bandit, pytest, golden-dataset regression checks and Jenkins-based deployment.
- Delivered XGBoost models across 1M+ records and 50+ locations, reusable AutoML components, MILP schedulers and real-time optimization frameworks.
- Partnered with business, engineering and DevSecOps teams to convert operating constraints into production ML and optimization services.

SELECTED PRODUCT WORK

Agentic Recruitment Platform

- Architected and developed the backend for a deployed product with six LangGraph agents, private Gemma 2B inference, workflow traceability and audit functions.

LEADERSHIP AND RECOGNITION

Recognition includes a Spot Award and Team Spot Award for real-time optimization delivery and promotion to Senior Consultant in July 2025.

EDUCATION

M.Sc. Mathematics | BITS Pilani | 2018 - 2022